Acceleration - Answer key

GCE Ordinary Level Physics - L acceleration - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Exercise 1 - A first acceleration

A taxi leaves the taxi rank at 3 m/s and reaches 13 m/s in 10 s on the avenue. Compute its acceleration in m/s^2.
Un taxi quitte la station a 3 m/s et atteint 13 m/s en 10 s sur l'avenue. Calcule son acceleration en m/s^2.

1. Corrige / Answer

Ex. 1 - a = delta_v / delta_t = (13 - 3) / 10 = 10 / 10 = 1 m/s^2. Le taxi gagne 1 m/s chaque seconde.

2. Exercise 2 - Convert first, then compute

A car goes from rest to 90 km/h in 6 s. Convert 90 km/h into m/s first, then find the acceleration.
Une voiture passe de l'arret a 90 km/h en 6 s. Convertis d'abord 90 km/h en m/s, puis calcule l'acceleration.

2. Corrige / Answer

Ex. 2 - 90 km/h = 90 / 3,6 = 25 m/s. a = delta_v / delta_t = (25 - 0) / 6 = 4,2 m/s^2 environ.

3. Exercise 3 - Braking, the negative case

A moto brakes from 15 m/s to a standstill in 3 s. Compute the acceleration and explain the minus sign.
Une moto freine de 15 m/s jusqu'a l'arret en 3 s. Calcule l'acceleration et explique le signe moins.

3. Corrige / Answer

Ex. 3 - a = (0 - 15) / 3 = -5 m/s^2. Le signe moins indique une vitesse qui diminue : c'est un freinage (deceleration).

4. Exercise 4 - From a speed-time graph

On a speed-time graph the line climbs from 2 m/s at t = 1 s to 10 m/s at t = 9 s. What is the acceleration?
Sur un graphe vitesse-temps, la ligne monte de 2 m/s a t = 1 s jusqu'a 10 m/s a t = 9 s. Quelle est l'acceleration ?

4. Corrige / Answer

Ex. 4 - Pente = (10 - 2) / (9 - 1) = 8 / 8 = 1 m/s^2.

5. Exercise 5 - Which one accelerates more?

Runner A goes from 2 m/s to 14 m/s in 6 s; runner B goes from rest to 15 m/s in 5 s. Which runner has the greater acceleration?
Le coureur A passe de 2 m/s a 14 m/s en 6 s ; le coureur B passe de l'arret a 15 m/s en 5 s. Quel est le plus fort en acceleration ?

5. Corrige / Answer

Ex. 5 - A : (14 - 2) / 6 = 2 m/s^2. B : (15 - 0) / 5 = 3 m/s^2. B accelere le plus vite, meme si B part de zero.

6. Exercise 6 - Braking distance in time

A truck loses speed at a steady -4 m/s^2 starting from 20 m/s. How many seconds does it take to stop completely?
Un camion perd sa vitesse a -4 m/s^2 constants depuis 20 m/s. Combien de secondes faut-il pour s'arreter completement ?

6. Corrige / Answer

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Ex. 6 - a = delta_v / delta_t donc delta_t = delta_v / a = (0 - 20) / (-4) = (-20) / (-4) = 5 s.

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